

Network-Level Threat Detection

Mapping and Destroying Attacker
Operations at Scale

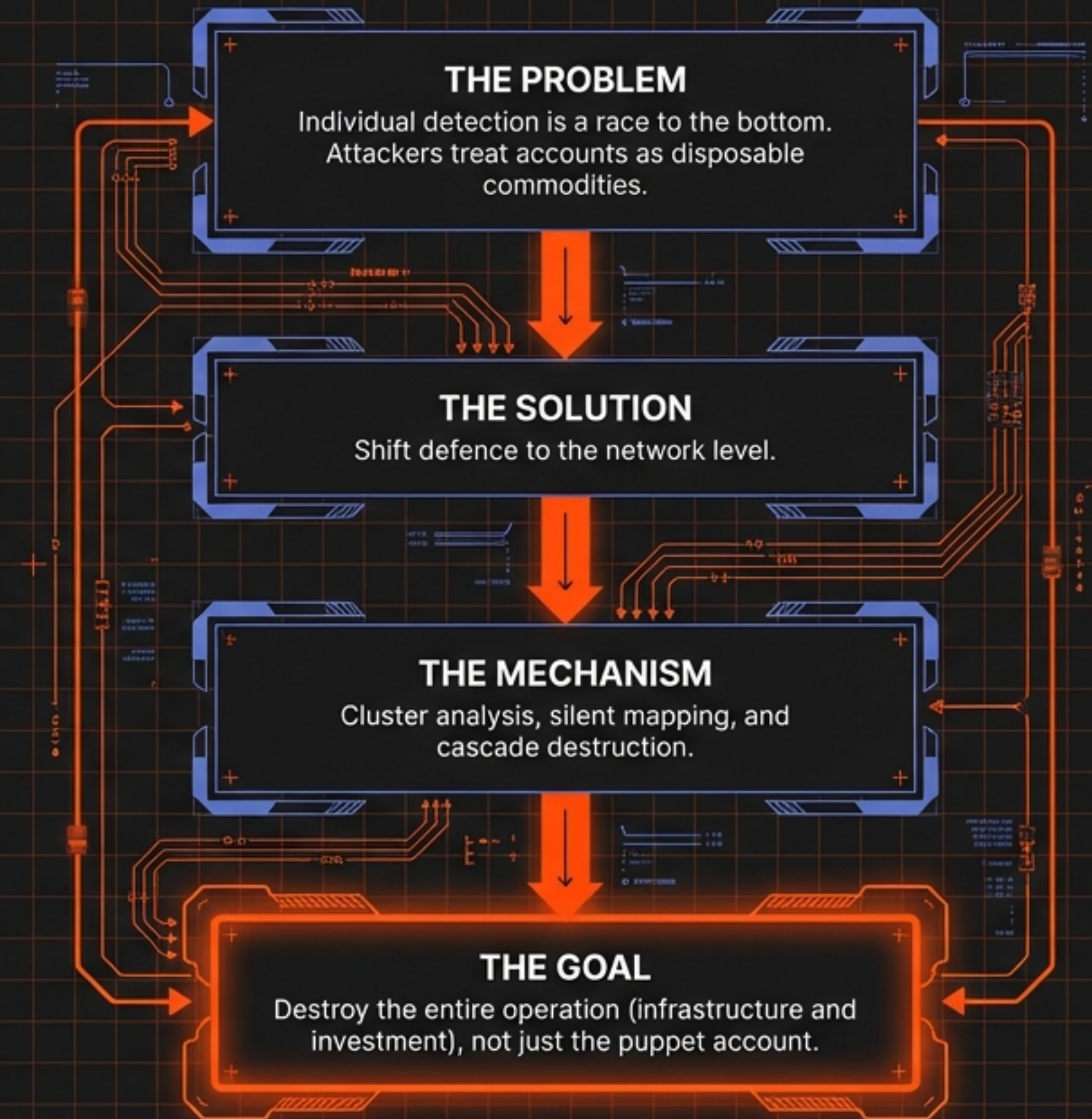
Version 0.6.14
Architecture Brief
25 Feb 2026

Architecture brief – threat detection and incident response at scale.

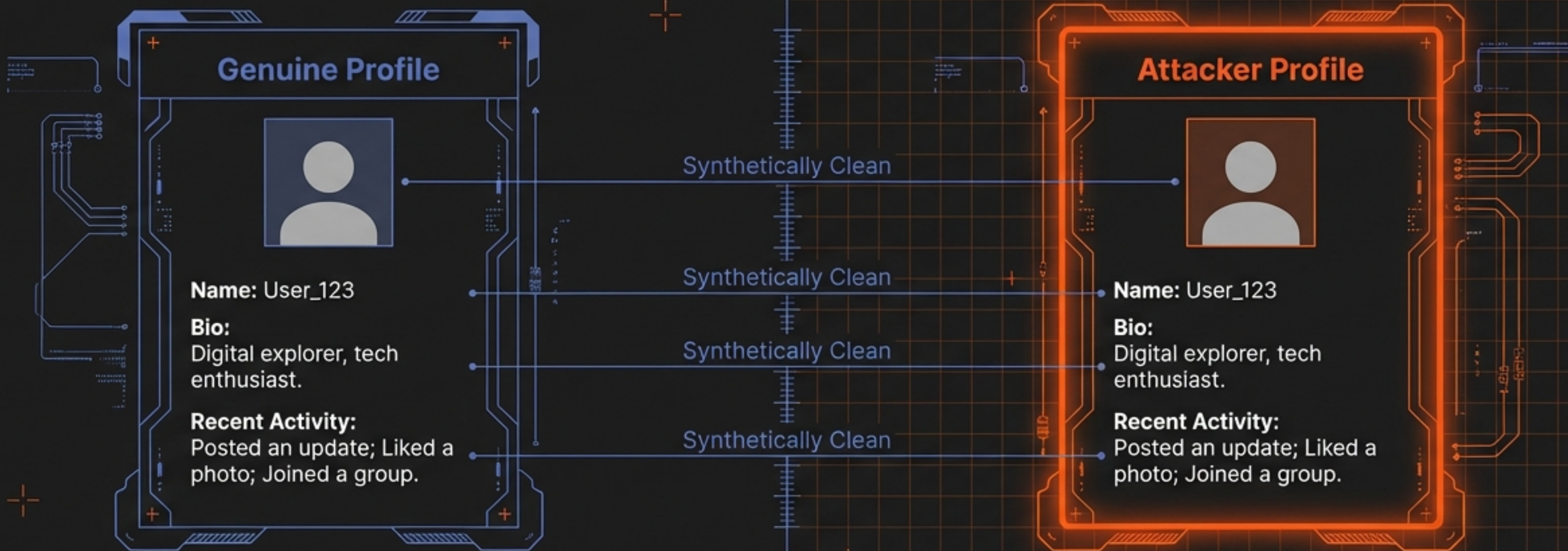
The Strategic Shift: From Individual Inspection to Graph Analysis

“It’s easy to create a single fake account. It’s nearly impossible to create thousands that don’t share patterns.”

STRATEGIC ANALYSIS // DATA POINT: NETWORK CORRELATION //
FORENSIC FOOTPRINT: ANOMALY DETECTION // SYSTEM: GRAPH THEORY



The Micro-Illusion: Why We Lose at the Individual Level

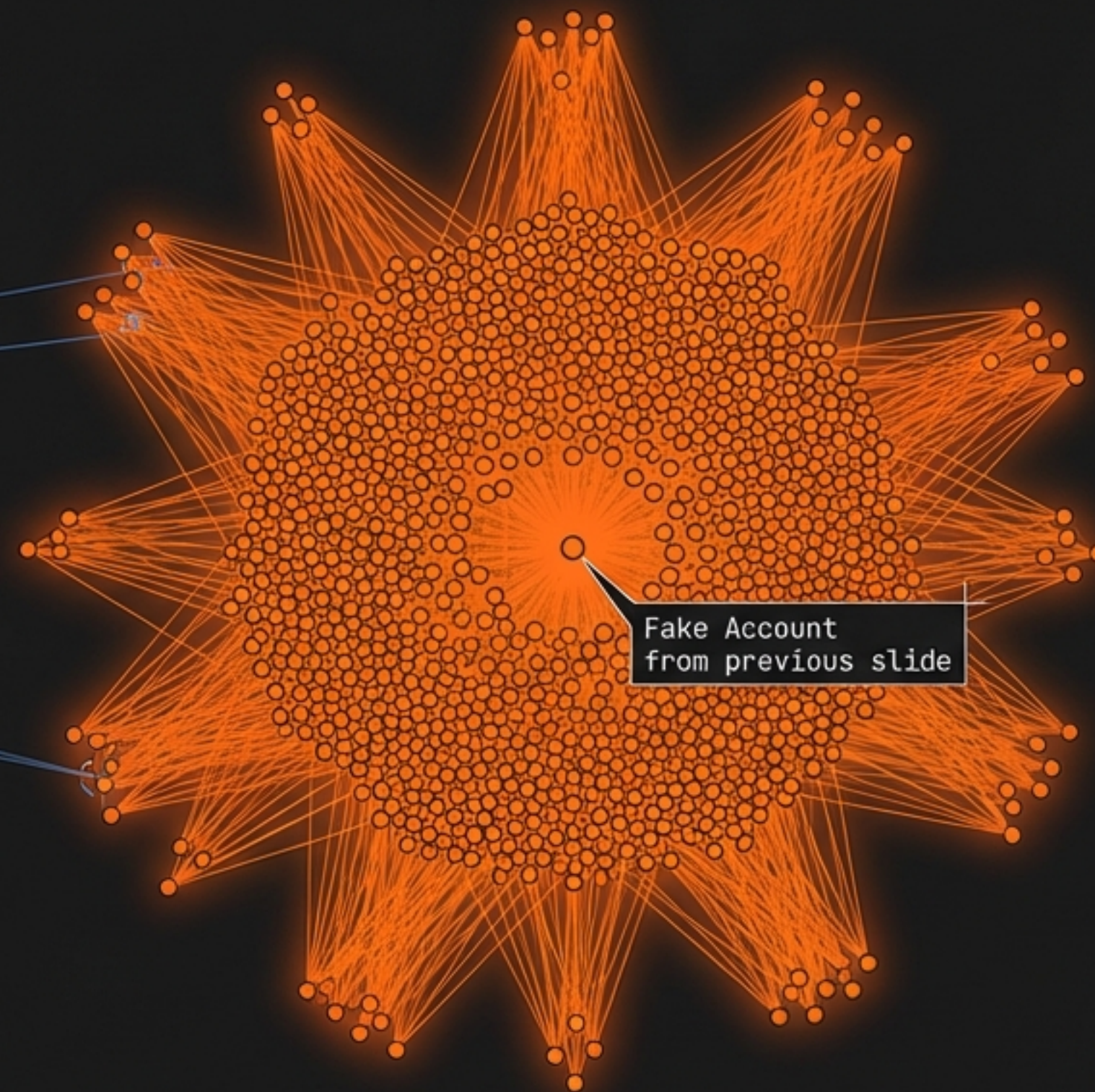


The Attacker's Strategy:

- **Trust Gaming:** Normal behavior, clean files, sleeper periods.
- **The Agentic Threat:** AI agents generate full conversations and plausible usage.
- **Result:** Manual detection finds no evidence of malice.

The Macro-Revelation: Scale Creates Patterns

Trust Web



IP Ranges

5,000 accounts need
5,000 IPs. Proxies
cluster around ASNs.

Timing

Batched creation.
Synchronized
sleep/wake cycles.

Topology

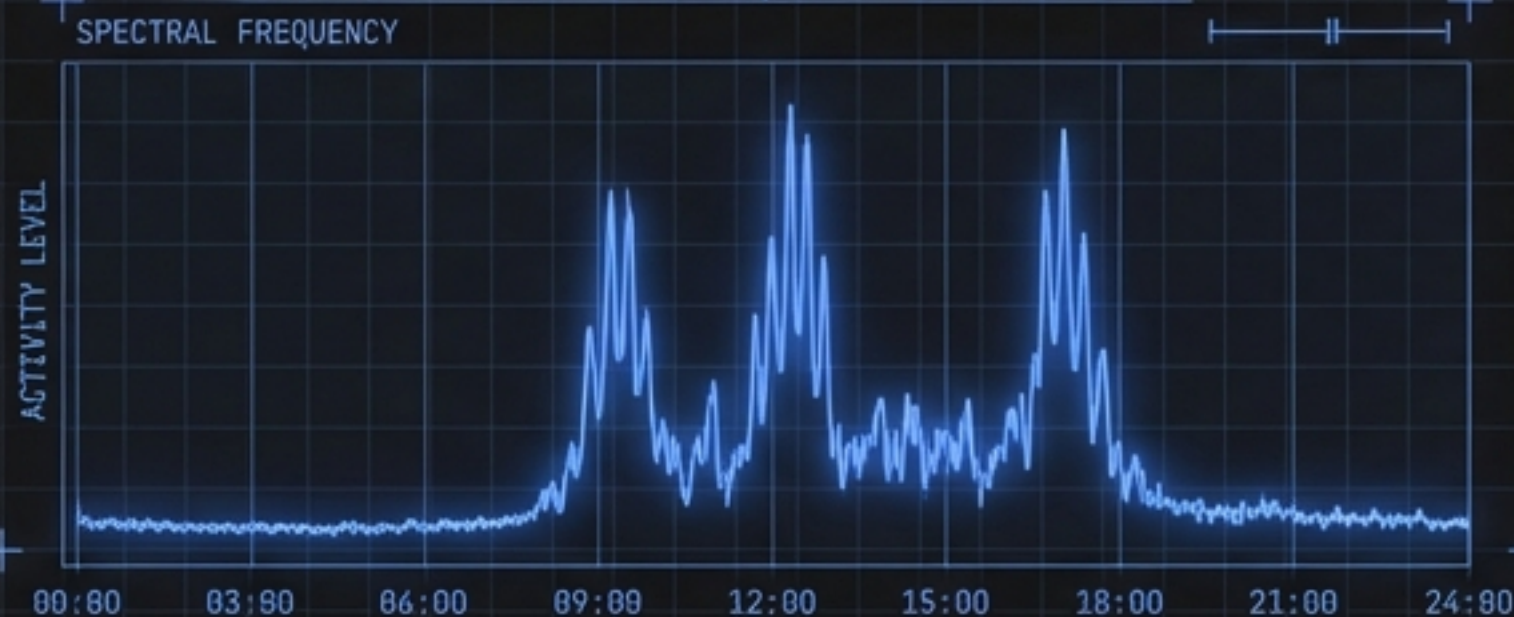
Insular clusters.
Few exits to
genuine graph.

Operating at scale always produces patterns that cannot be hidden.

The Paradox of Randomness

High entropy is a fingerprint.

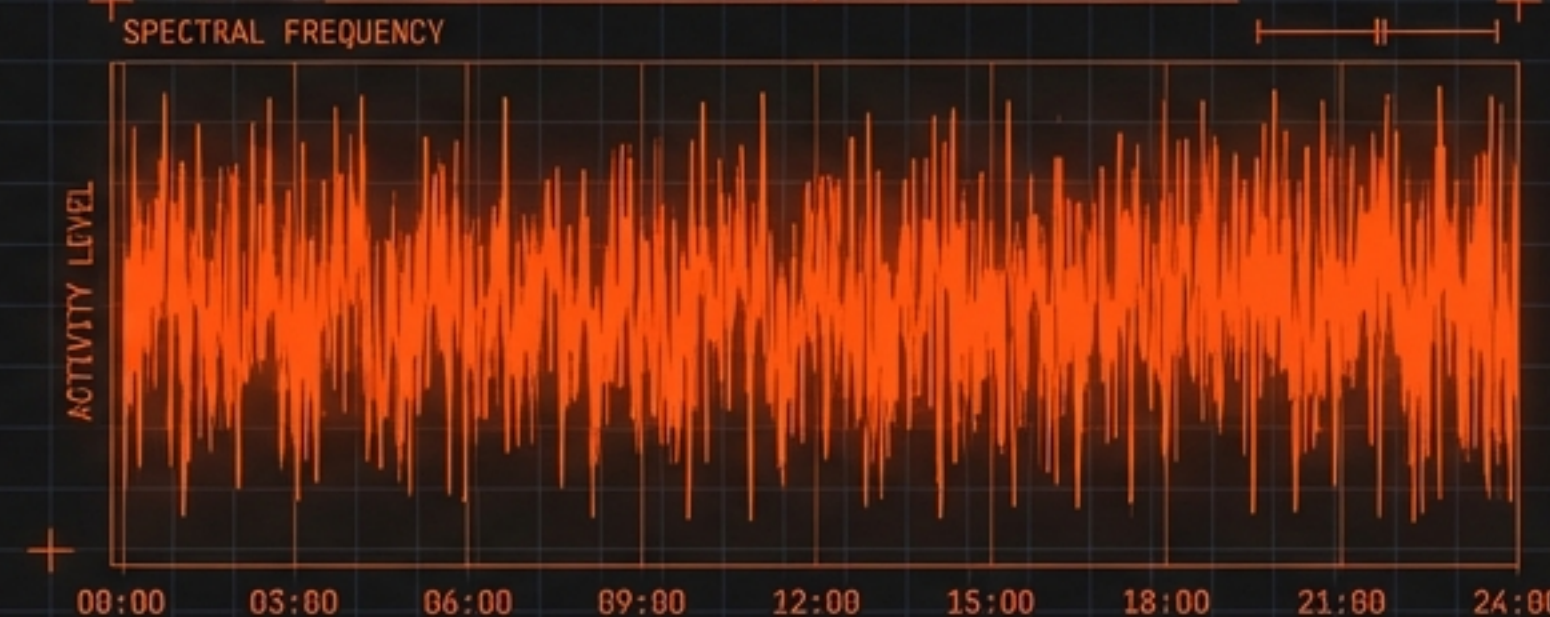
Genuine User



DATA ANNOTATION: (JetBrains Mono)
ANALYTICS: Predictable habits. Low entropy.

ENTROPY SCORE:
2.4 (NORMAL)

Attacker Script

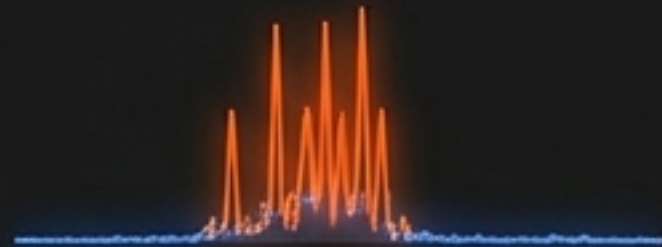
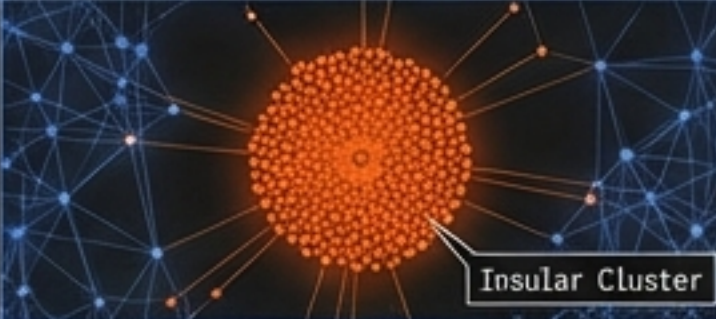


DATA ANNOTATION: (JetBrains Mono)
ANALYTICS: Pure randomization. High entropy.

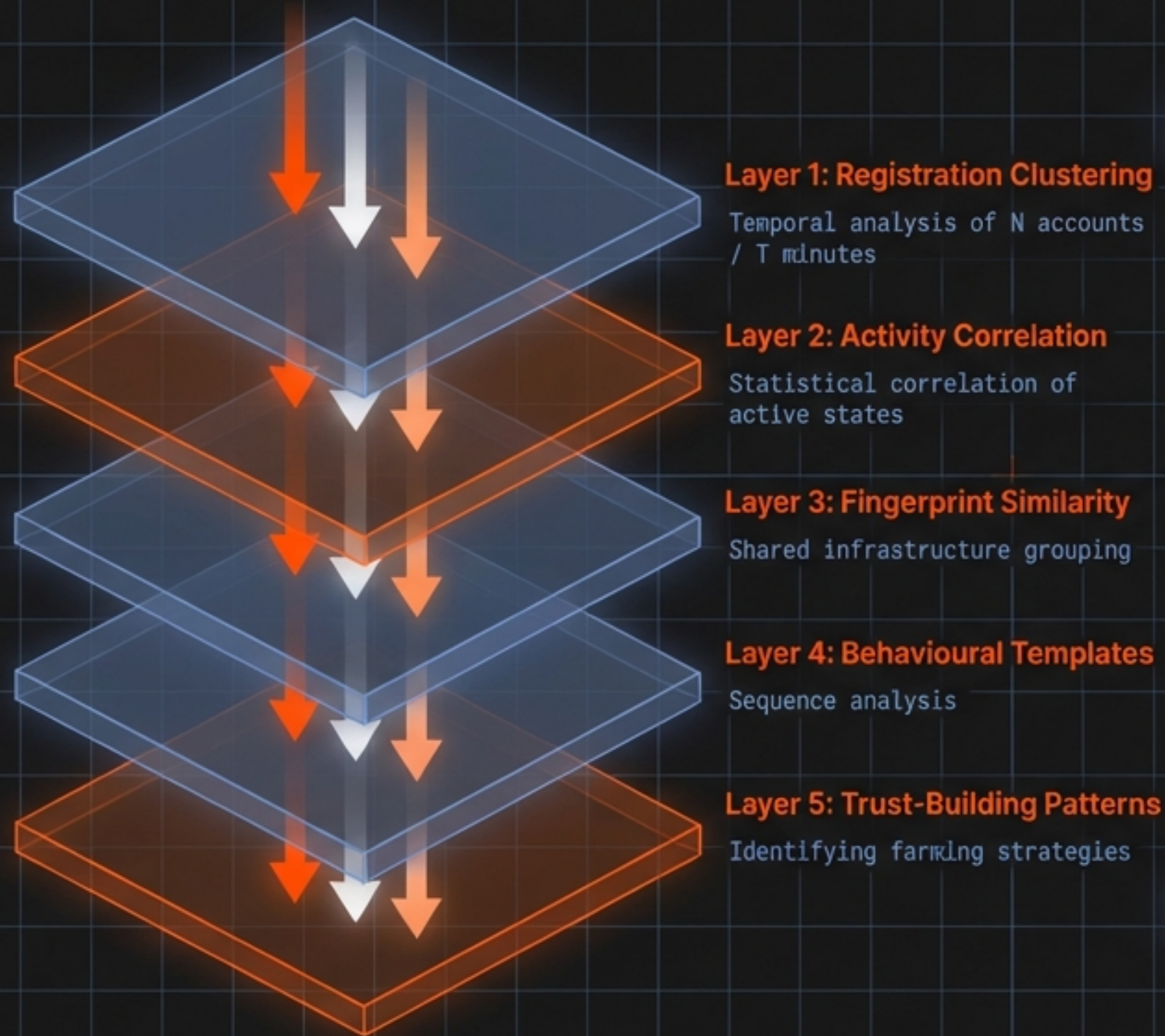
ENTROPY SCORE:
9.8 (CRITICAL)

INSIGHT: A cluster of accounts with high-entropy behaviour is suspicious precisely because it is too random. 'Randomised' behaviour shares the fingerprint of the randomisation algorithm.

Hard to Hide: The Signals of Synthetic Scale

SIGNAL TYPE	DETAIL ANALYTICS	DISTRIBUTION ANOMALIES
Device/Browser Fingerprints	Shared screen resolutions, WebGL renderers, timezones, and language configs across thousands of instances.	
Transaction Patterns	Statistical distributions of file sizes, types, and sending frequency that differ from organic usage.	
Registration Patterns	Correlations in email domains, username syntax, and onboarding velocity.	
Network Topology	Dense clusters with few exits to the genuine graph.	

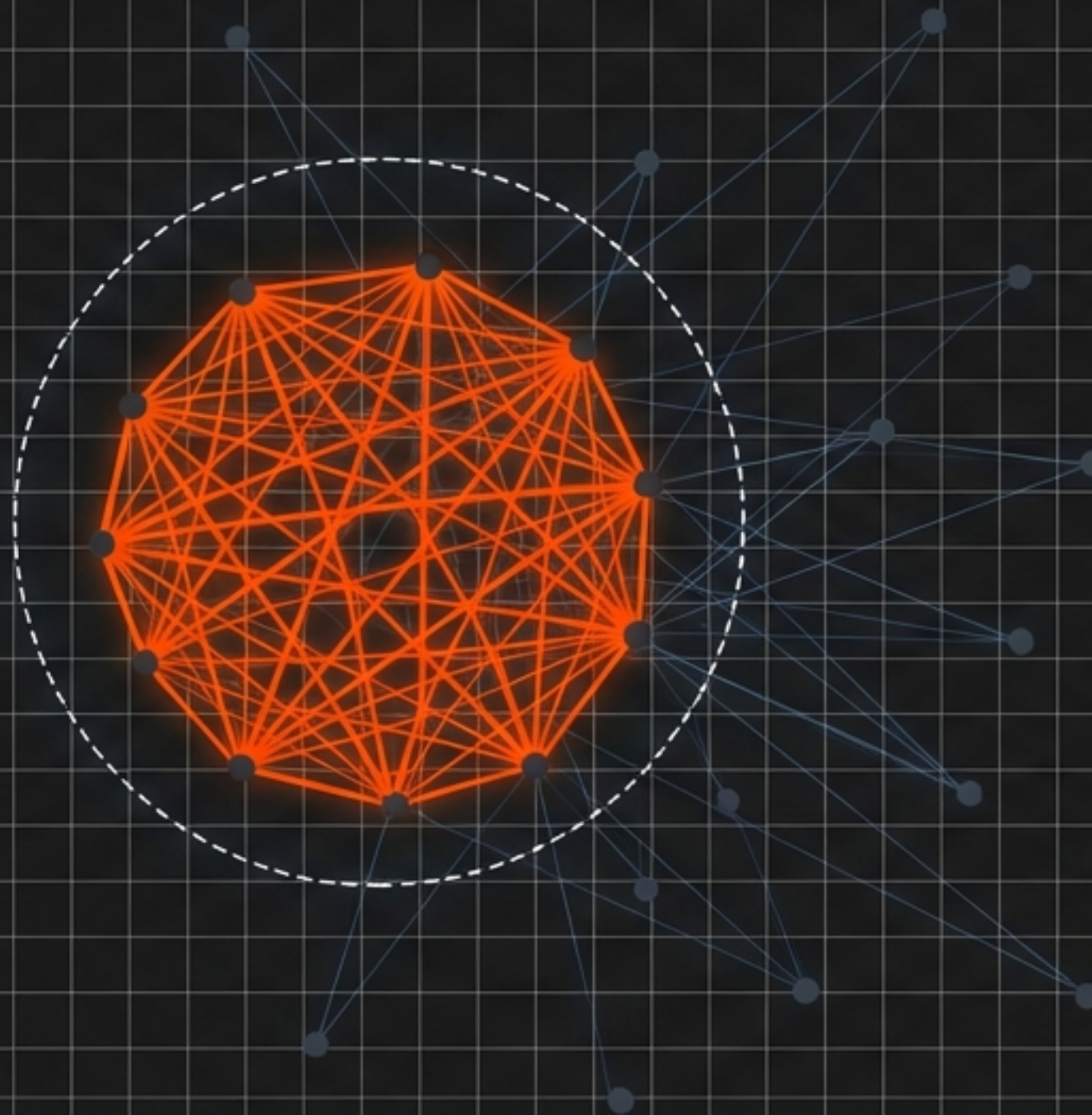
The Detection Engine: Layered Cluster Analysis



Key Shift: From “Is this account malicious?” to “Does this group share coordinated patterns?”

Graph Approach & Community Detection

The Technical Definition:
The attacker network
appears as a dense
subgraph with
disproportionately few
connections to the
genuine user
population.



Algorithms:

- Louvain Method
- Label Propagation
- Spectral Clustering

Silent Detection: The Intelligence Option

Why we map the network before pulling the trigger.

● REC



STRATEGIC BENEFITS

1. **MAP THE FULL NETWORK:** Identify more infrastructure over time.
2. **TRACE COMMAND & CONTROL:** Learn the *How*, not just the *That*.
3. **MAXIMIZE DESTRUCTION:** Destroy the entire operation in one stroke.

Decision Matrix: When to Wait, When to Act

FORENSIC ARCHITECTURAL // ANALYTICAL FRAMEWORK V2.4 // CONFIDENTIAL MATRIX

ANALYTICAL MATRIX

Scenario: Active Harm (Malware)

Action: ACT IMMEDIATELY

Priority: Protect users.

Scenario: Suspected Network

Action: WAIT & MAP

Priority: Build the case.

Scenario: Low Trust / Suspicious

Action: MONITOR

Protocol: Reduce trust scores,
do not tip off.

Scenario: High Confidence

Action: PREPARE TAKEDOWN

Protocol: Coordinate simultaneous
destruction.

Cascade Destruction: The Nuclear Option

The decisive action to neutralize the attacker's entire operation.



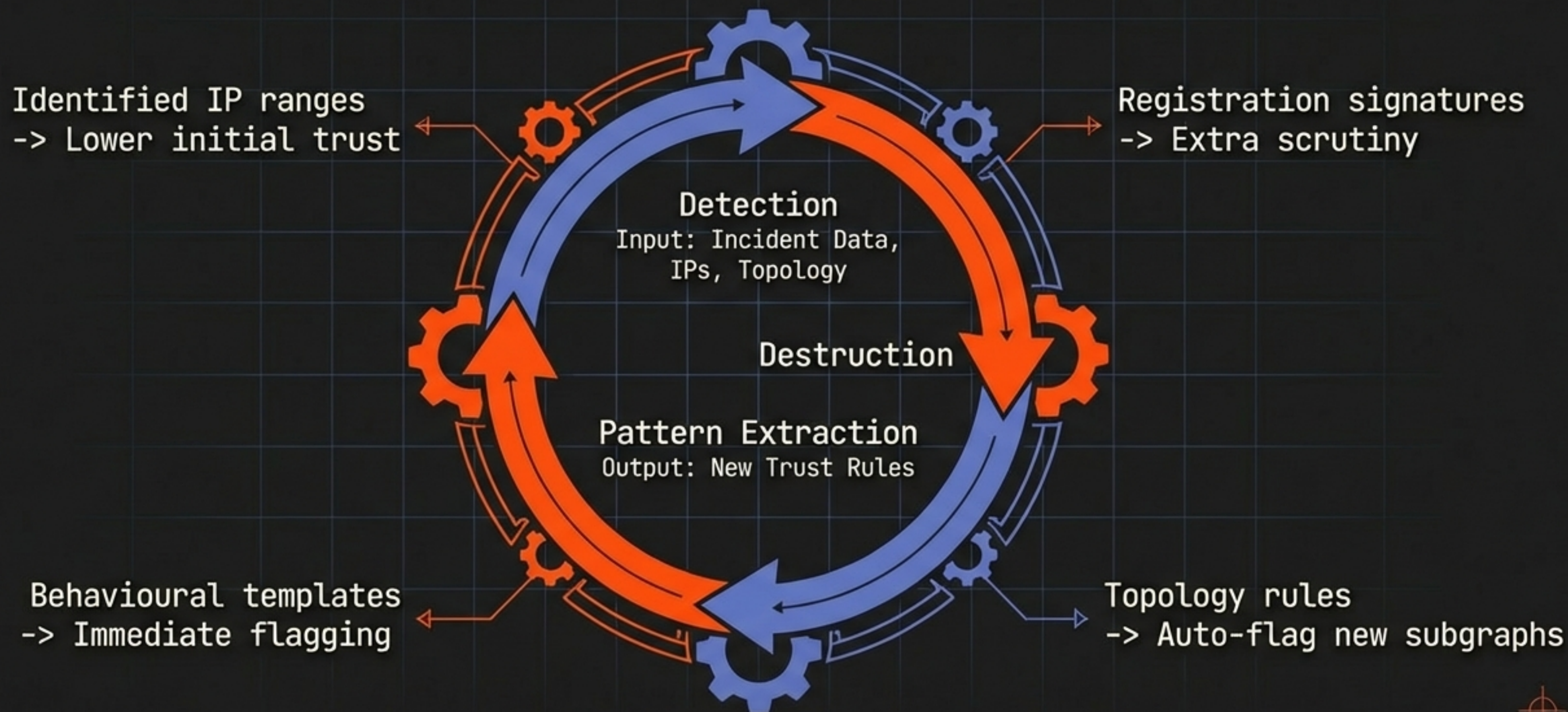
COST TO THE ATTACKER

1. **Infrastructure Burned:** IP ranges and browser configs useless.
2. **Investment Lost:** Weeks of trust-building time destroyed instantly.
3. **Secrecy Lost:** C&C structure mapped.
4. **Future Access Denied:** Pattern signatures extracted.

“Turning a minor incident into a major defensive win.”

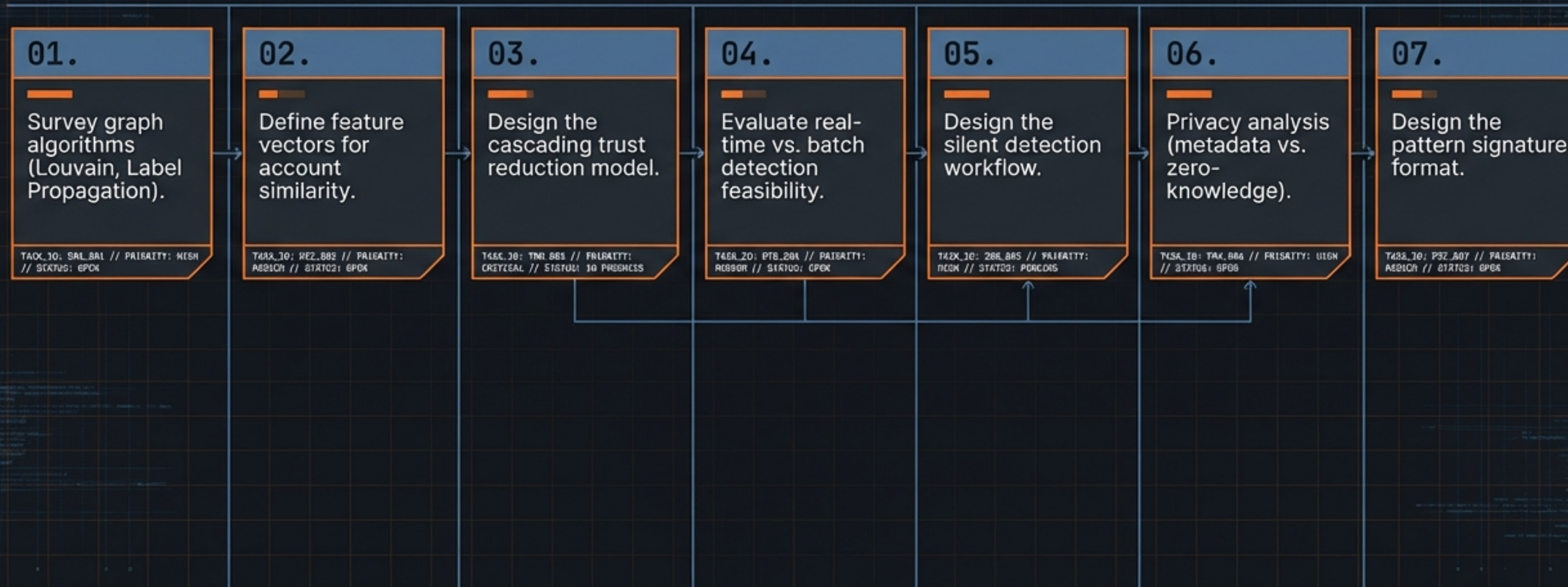
The Learning System: Feeding Back into the Trust Web

FORBNSIC ARCHITECTURAL // ANALYTICAL FRAMEWORK V2.4 // LEARNING LOOP VISUALIZATION



AppSec Research & Implementation Tasks

FORENSIC ARCHITECTURAL // ANALYTICAL FRAMEWORK V2.4 // CONFIDENTIAL PROJECT PIPELINE



Acceptance Criteria & Definition of Done

- [] Trust violation triggers network mapping.
- [] Cluster detection identifies pattern-sharing accounts.
- [] Cascading trust reduction applied.
- [] Historical file review triggered.
- [] Affected recipients notified.
- [] Pattern signatures fed back to detection.
- [] Silent detection mode operational.
- [] Coordinated network destruction capability live.
- [] Incident management framework updated.